

17 Ans: A

Zero intensity => node detected

At distance 0.34 m => 2nd Harmonics Standing Waves for a Node to fit ,
=> 0.34 m = $\frac{1}{4}$ wavelength, Wavelength = 1.36 m

$$\text{Initial freq } f = 340/1.36 = 250 \text{ Hz}$$

The next higher frequency to have a Node at 0.34 m is when

$$0.34 \text{ m} = \frac{3}{4} \text{ wavelength} , \text{ Wavelength}' = \frac{4}{3} (0.34) = 0.45333 \text{ m}$$

$$\text{New freq } f' = 340/0.45333 = 750 \text{ Hz}$$